

Data & AI Case Study

Mapped to two Wuerth internship roles

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This is my independent case study. I mapped the requirements of two Wuerth Data & AI internship postings to work I have already built and measured in my own portfolio (now 23 repositories, including the decision-chain integration capstone -- one real dataset through the whole distributor chain with 13 machine-checked reconciliation identities plus four additive ones -- an MCP agentic-integration server with contract-enforced result provenance and idempotent result caching, two warehouse-logistics flagships, a supply-network optimizer, an energy forecasting + dispatch study, a visual quality-inspection study with SPC monitoring, a fraud-operations study with gated retrain promotion, a predictive-maintenance policy study, and a live installable

Job #1 (Agentic) Automation with Low-code Platforms
This explains PWA. The work includes a live installable MCP agentic-integration server on synthetic data (except retail-analytics-real and decision-chain, measured and labelled on real public data) and exist to prove the methods work and are measurable -- not to forecast Wuerth outcomes. Workflows, low-code (No / Power Automate), connecting systems & APIs, document automation, ROI, rapid prototyping.

Job #2 -- Data & AI Analytics

BI / Power BI, KPI dashboards, forecasting & predictive analytics, Python & SQL, data modelling, turning data into decisions.

Who Wuerth is (public info)

Wuerth (public profile, approximate): a large family-owned German-HQ group in assembly/fastening and industrial MRO distribution -- ~400+ companies in 80+ countries, ~87,000 employees, ~EUR 20B+ revenue, a catalogue in the millions of articles, multi-channel (field sales, branches, e-commerce, e-procurement/EDI, ORSY inventory systems). A business that runs on assortment, pricing, availability, logistics, and high-volume transactional documents -- a natural fit for applied Data & AI.

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Opportunity map (summary)

Wuerth business area (public-info) -> Data/AI approach -> repo + measured result (synthetic unless noted)

Area (public-info)	Approach	Repo	Result (synthetic -- labelled if real)
Procurement / assortment / cross-sell	MILP vs greedy; Apriori/FP-growth; rule redundancy pruning	revops-optimizer, market-basket-analysis	captured 2.1x revenue of value-added uplift; the H49 kept stateless information MASE 0.38; MASE 0.987 / RMSE 0.918 (naive + k-fold cross-validated)
Pricing & margin	Elasticity + leakage; endogeneity fix; per-move robustness gate; ablation	revops-optimizer, sales-kpi-analytics, ml-models-lab	MAE 0.127421; working capital (cycle+safety exactly), MASE turns 0.99% diff; pacing files 10 real lead times 668, 89.4% yearly stock-out-year with check/lookup above the band, reported; rep standard deviation 0.92% of the targeted table; 5% head-of-trail headroom
Inventory / replenishment	Newsvendor + forecast + ROP/EOQ policy + supplier reliability	distributor-intel-platform, ml-models-lab	MAE 0.127421; working capital (cycle+safety exactly), MASE turns 0.99% diff; pacing files 10 real lead times 668, 89.4% yearly stock-out-year with check/lookup above the band, reported; rep standard deviation 0.92% of the targeted table; 5% head-of-trail headroom
Sales KPIs & analytics	KPIs + rolling-origin CV + pacing interval + rep decomposition	sales-kpi-analytics	MAE 0.127421; working capital (cycle+safety exactly), MASE turns 0.99% diff; pacing files 10 real lead times 668, 89.4% yearly stock-out-year with check/lookup above the band, reported; rep standard deviation 0.92% of the targeted table; 5% head-of-trail headroom
Logistics / routing	OR-Tools CP-SAT VRP + robustness + Fleet Size and Mix VRP	route-optimizer	MAE 0.127421; working capital (cycle+safety exactly), MASE turns 0.99% diff; pacing files 10 real lead times 668, 89.4% yearly stock-out-year with check/lookup above the band, reported; rep standard deviation 0.92% of the targeted table; 5% head-of-trail headroom
E-procurement automation	Agentic + n8n low-code + reliability benchmark + content-verification layer + dry-run flow cost estimator	agentic-automation-lab, agent-flow-studio	MAE 0.127421; working capital (cycle+safety exactly), MASE turns 0.99% diff; pacing files 10 real lead times 668, 89.4% yearly stock-out-year with check/lookup above the band, reported; rep standard deviation 0.92% of the targeted table; 5% head-of-trail headroom
Document processing	Agentic extract + 7-rule validation gate + cost model	doc-extract-agent	MAE 0.127421; working capital (cycle+safety exactly), MASE turns 0.99% diff; pacing files 10 real lead times 668, 89.4% yearly stock-out-year with check/lookup above the band, reported; rep standard deviation 0.92% of the targeted table; 5% head-of-trail headroom
Customer retention	Decline + churn classifiers + WMS twin + factory/plant sim (Story Mode, 894-element plant, definable objects, multi-way line sim + fluids solver); slotting + DES + packing + pick-path routing + order batching	revops-optimizer, ml-models-lab	MAE 0.127421; working capital (cycle+safety exactly), MASE turns 0.99% diff; pacing files 10 real lead times 668, 89.4% yearly stock-out-year with check/lookup above the band, reported; rep standard deviation 0.92% of the targeted table; 5% head-of-trail headroom
Warehouse / WMS (new)		logistics-flow-studio (WarehouseTwin v3.19), logistics-digital-twin (engine)	MAE 0.127421; working capital (cycle+safety exactly), MASE turns 0.99% diff; pacing files 10 real lead times 668, 89.4% yearly stock-out-year with check/lookup above the band, reported; rep standard deviation 0.92% of the targeted table; 5% head-of-trail headroom
Supply-network design (new)	Facility MILP + flows + safety stock + service frontier + growth plan	supply-network-opt	MAE 0.127421; working capital (cycle+safety exactly), MASE turns 0.99% diff; pacing files 10 real lead times 668, 89.4% yearly stock-out-year with check/lookup above the band, reported; rep standard deviation 0.92% of the targeted table; 5% head-of-trail headroom
Energy management / facilities (new)	Load forecast (rolling CV) + peak-shaving LP + causal dispatch backtest + degree-day decomposition	energy-demand-forecast	MAE 0.127421; working capital (cycle+safety exactly), MASE turns 0.99% diff; pacing files 10 real lead times 668, 89.4% yearly stock-out-year with check/lookup above the band, reported; rep standard deviation 0.92% of the targeted table; 5% head-of-trail headroom
Quality inspection (new)	Clean-only anomaly detection; pre-registered rule; SPC p-chart + WE rules; measured OCAP	quality-anomaly-vision	MAE 0.127421; working capital (cycle+safety exactly), MASE turns 0.99% diff; pacing files 10 real lead times 668, 89.4% yearly stock-out-year with check/lookup above the band, reported; rep standard deviation 0.92% of the targeted table; 5% head-of-trail headroom
Real data (completed)	Cleaning + RFM + returns + lifecycle + leakage-safe CV	retail-analytics-real	MAE 0.127421; working capital (cycle+safety exactly), MASE turns 0.99% diff; pacing files 10 real lead times 668, 89.4% yearly stock-out-year with check/lookup above the band, reported; rep standard deviation 0.92% of the targeted table; 5% head-of-trail headroom
Chain integration & reconciliation (new)	Provenance-tagged pipeline + identity ledger; MCP agentic layer w/ machine-readable result provenance + idempotent caching	decision-chain, chain-mcp	MAE 0.127421; working capital (cycle+safety exactly), MASE turns 0.99% diff; pacing files 10 real lead times 668, 89.4% yearly stock-out-year with check/lookup above the band, reported; rep standard deviation 0.92% of the targeted table; 5% head-of-trail headroom
Fraud & transaction-risk ops (new)	Cost-based alerting; gated retrain promotion; selective-labels feedback sim	fraud-detection-ops	MAE 0.127421; working capital (cycle+safety exactly), MASE turns 0.99% diff; pacing files 10 real lead times 668, 89.4% yearly stock-out-year with check/lookup above the band, reported; rep standard deviation 0.92% of the targeted table; 5% head-of-trail headroom
Maintenance & asset reliability (new)	Censored Weibull + age-replacement vs condition-based + CBM re-tuning	predictive-maintenance	MAE 0.127421; working capital (cycle+safety exactly), MASE turns 0.99% diff; pacing files 10 real lead times 668, 89.4% yearly stock-out-year with check/lookup above the band, reported; rep standard deviation 0.92% of the targeted table; 5% head-of-trail headroom

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Job #1 -- (Agentic) Automation with Low-code

Every posting requirement -> repo + artifact + measured proof (all synthetic data).

Requirement	Repo / artifact	Proof (synthetic data only)
Build AI-agent workflows -- and measure how they fail	agentic-automation-lab (+ eval/reliability.py benchmark + eval/verification.py content-verification layer)	agentic tool-use loops; reliability benchmark (1,350 seeded trials): content faults fail silently past the guards and retries can't zero them (assumed fault rates, stated); contract verification never fails; the same fault schedule: delivered-correct 88.7% -> 98.8%, silent-wrong 11.523/yr or real engines as available models, 260 typed schemas; never-crash errors; contract-enforced, machine-readable result provenance (data label + engine commit + determinism flag); idempotency + result caching -- the key binds args + the exact engine checkout, and every success says computed-or-replayed (provenance.cache, contract-required); 193 tests incl. live JSON-RPC handshake
Agentic integration via an open standard (MCP)	chain-mcp (official mcp Python SDK; Claude Desktop / Code configs)	
Low-code (n8n / Power Automate)	agentic-automation-lab: n8n/rfq_intake_agent.json	RFQ-intake agent: ~EUR625k/yr flow builder; ~EUR47k/yr; dry-run cost/latency estimator: declared rate card, branch scenarios enumerated and proven against real engine runs branch-for-branch -- 'emphatically not a bill' (78 node tests)
Visual / flow-based building	agent-flow-studio (+ estimator.js dry-run estimator)	
Connecting systems / APIs	agentic-automation-lab connector layer	agents call external APIs
Process automation (back-office)	doc-extract-agent	RFQ/invoice -> structured extract + validate; ~EUR145k/yr; combined-gate precision 70.0% -> 87.5%; cost model: auto-post pays only above 98.4% precision -- the gate alone would lose money (modeled), the pre-fill carries the value
Document intake / understanding	doc-extract-agent + 7-rule validation layer + cost model (eval/run_cost)	
Rapid prototyping	agent-flow-studio, agentic-automation-lab, logistics-flow-studio	hand-written simulator ~300 lines, zero deps; lessons incl. superdense coding (no-PH, demo -- the encoded pair is installable, single-file, 40k/yr, and factory/plant simulator). (BWA physics harnesses + 53 self-test + 93 structural checks pass; BioHash: mid-range wins from a 10x smaller circuit -- negatives kept (both tips post, no memory-frontier point, corpus-dependent)
Prototyping + education (hype-free)	quantum-explainer -- LIVE: dimitres-kisimov.github.io/quantum-explainer bio-efficient-ai (learned BioHash vs random, public MNIST benchmark)	
Show automation ROI	automation-roi-explorer	EUR383k/yr net modelled
Python engineering	all automation repos	Python throughput
Stakeholder communication	automation-roi-explorer	business-readable ROI views

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Job #2 -- Data & AI Analytics

Every posting requirement -> repo + artifact + measured proof (synthetic data unless table 159.0% a Github 665 (model shape,

Requirement	Repo / artifact	Proof (synthetic unless noted)
One dataset through the whole chain (real data + labelled layers)	decision-chain (run artifact + offline dashboard; FAIL path per identity)	KPI layer + exec PDF; pacing EUR10.91M = 95.7% of plan, 80% interval; back-check: realised EUR11.28M landed above the band; no honest miss; rep 100% pass; page report 1 Monte-Carlo 1000000 iterations; under 1% DAX standardization); 95% of the line; require 10% dispersion for lead time; portfolio tanks suppliers; price-move gate: 17/29 ACCEPT, the biggest move HELD (67% carry) -- a screen, not a guarantee; bootstrap skill CI = 0.457 [0.381, 0.528]; ablation (freq. slope knockout -0.247 PR-AUC, endogeneity-control knockout +1.446 RMSE; the calibration knockouts Mfr-identical CPE-ABD-fee-kept honest exception; null knockouts not counted as wins (numpy only, CI-reproducible)
BI / Power BI	revops-optimizer: powerbi/DAX_measures.md + kpi_uplift_risk	
KPI dashboards / metrics + fair performance measurement	sales-kpi-analytics + pacing bullet chart + repperf.py	
Uncertainty / risk quantification	revops-optimizer simulate.py + robustness.py; ml-models-lab BENCHMARK_CI.md	
Inventory policy / working capital	distributor-intelligence-platform dip/inventory.py + dip/reliability.py; supply-network-opt frontier	
Predictive analytics / forecasting	sales-kpi-analytics; ml-models-lab global forecaster	
Forecasting rigour + reconciliation guard	distributor-intelligence-platform (MRO command center)	
Energy forecasting + optimization	energy-demand-forecast (forecasters + LP + dispatch backtest + degree-day decomposition; 72 tests)	
Python for analytics	sales-kpi-analytics, revops-optimizer	end-to-end pipelines
SQL / data modelling	sales-kpi-analytics SQL set	
Excel-level tabular analysis	sales-kpi-analytics	
Optimization / prescriptive	revops-optimizer; supply-network-opt; logistics-digital-twin	
Elasticity / statistical modelling	revops-optimizer; ml-models-lab (+ ablation -> docs/ABLATION.md, drift-gated)	
Data into business decisions	sales-kpi-analytics; market-basket-analysis	
Market-basket / cross-sell	market-basket-analysis + affinity.py + redundancy.py	
Classification / anomaly detection	ml-models-lab; fraud-detection-ops	
Model lifecycle / MLOps (gated retrain)	fraud-detection-ops challenger.py + feedback.py	
Reliability / maintenance (survival)	predictive-maintenance pdm/policy.py + pdm/cbm_tuning.py	
Visual quality inspection (vision)	quality-anomaly-vision (3 detectors, pre-registered rule; SPC + OCAP; 59 tests)	
Warehouse / intralogistics + factory / process industry	logistics-flow-studio (WarehouseTwin v3.19); logistics-digital-twin (engine)	
Logistics / ops analytics	route-optimizer (46 tests, fleetmix.py); supply-network-opt (71 tests, growth.py)	
Real, messy data (completed)	retail-analytics-real (UCI Online Retail II, 1,067,371 real rows, CC BY 4.0)	
Present to decision-makers	revops-optimizer exec deck	exec narrative deck